E.2 SELECTION BIAS OF LLMS

In experiments illustrated in table 1, we observe a selection bias of LLaMA-65B. We show the results for each class in Table 4. Note that we randomly permute the test set to make sure all questions are balanced with [A,B,C,D].

First, we see that the zero-shot performances in different class are quite different. Class 'A' gets an accuracy as high as 71.58% and class 'D' only gets 52.28%. Second, with more demonstrations, the overall accuracy is improved, from 60.61% to 63.16%, demonstrating the effectiveness of ICL. However, the accuracy of class 'A' largely decreases, while the accuracies for 'B', 'C', and 'D' all increase. The above findings indicate that LLaMA-65B has a selection bias in both zero-shot and few-shot scenarios.

Table 4: **Accuracies for each class on our balanced MMLU.** We use the LLaMA-65B and vary the number of demonstrations from 0 to 5.

Class	0	1	2	3	4	5
A	71.58	57.31	58.25	57.31	56.96	54.74
B	59.65	66.90	69.94	70.41	69.59	70.76
C	58.95	67.02	65.50	64.80	66.78	67.60
D	52.28	59.65	60.00	59.30	59.88	59.53
Avg.	60.61	62.72	63.42	62.95	63.30	63.16